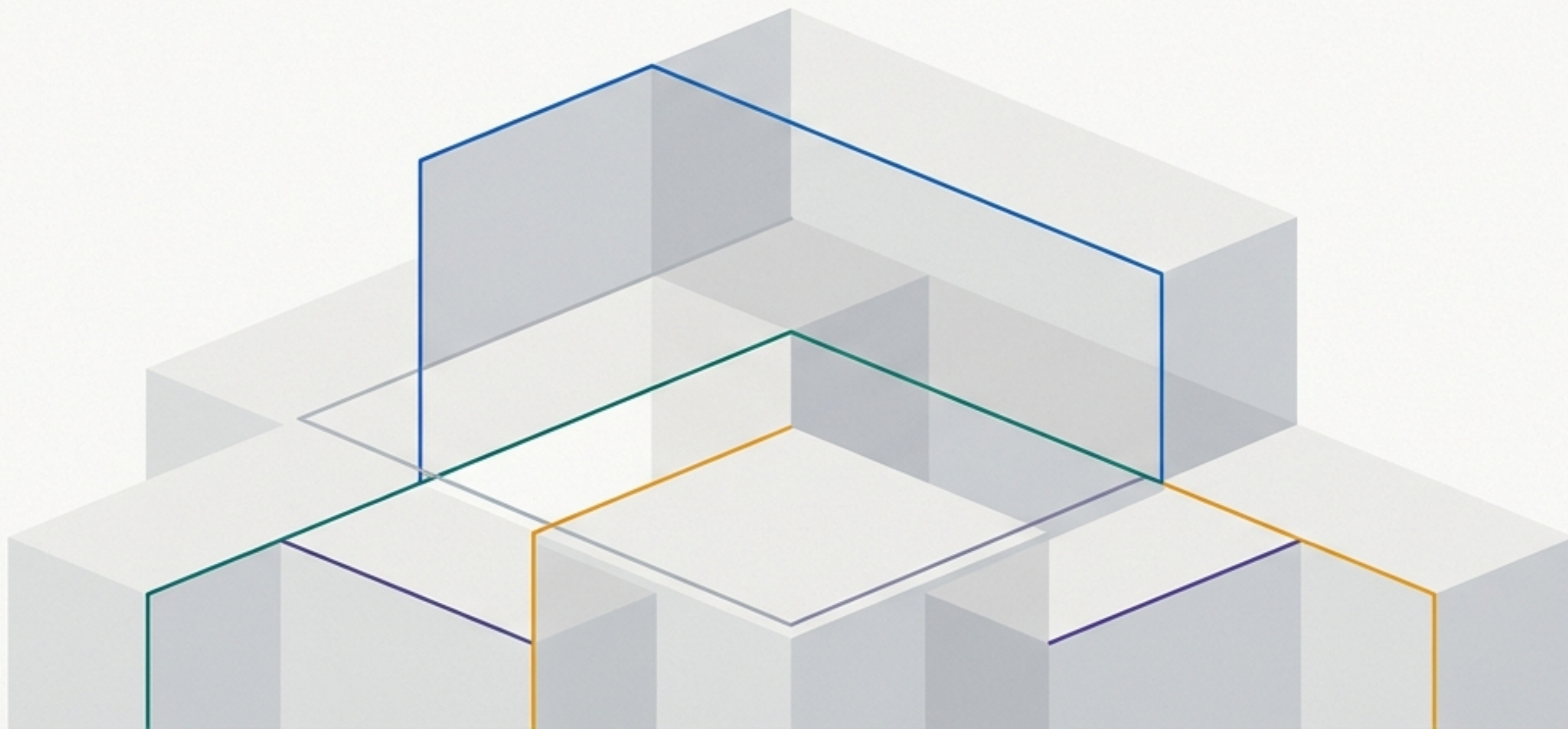




Building the Modern Enterprise

A unified platform for resilience, automation, and security, from endpoint to cloud.



The Four Pillars of the Modern Enterprise Platform

Google provides a complete, integrated platform for building the modern enterprise—one that is resilient, automated, and secure from its foundational infrastructure to the end-user's browser. This platform is built upon four interconnected strategic pillars.



Pillar 1: The Modernized Core

Focus: Infrastructure Migration & Business Resiliency



Pillar 2: The Automated Engine

Focus: Operational Orchestration & Application Integration



Pillar 3: The Secure Perimeter

Focus: Endpoint, Network & Data Security



Pillar 4: The Foundation of Trust

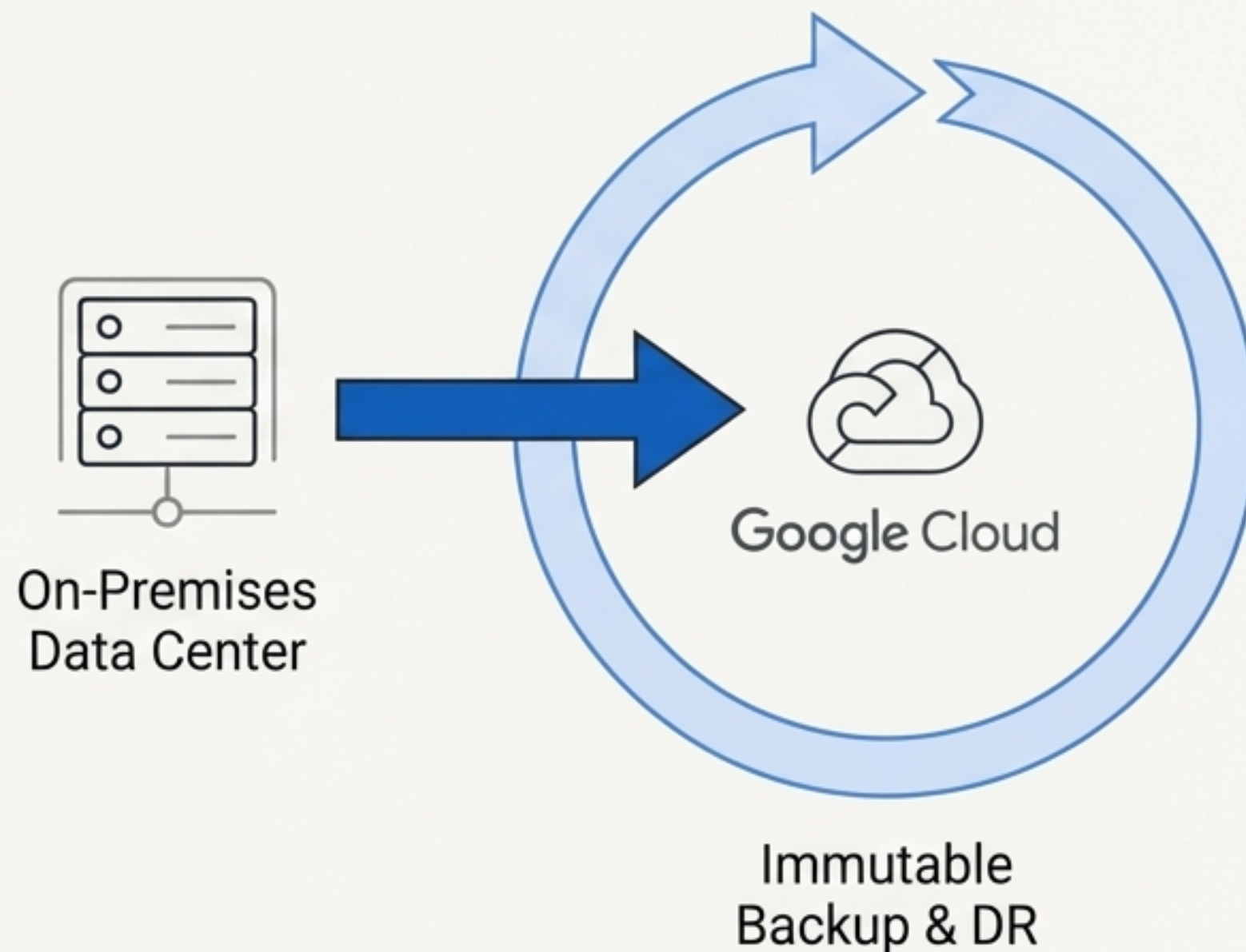
Focus: Verified Compliance & Developer Enablement

Pillar 1: The Modernized Core

De-risking the move to the cloud and ensuring absuring absolute business continuity.

The Challenge: Enterprises face a dual challenge: transitioning thousands of interdependent legacy workloads with minimal downtime and protecting those assets against catastrophic outages or ransomware attacks once in the cloud.

The Google Solution: A unified approach that combines a data-driven migration factory with managed, immutable backup and disaster recovery services.





Pillar 1: The Modernized Core

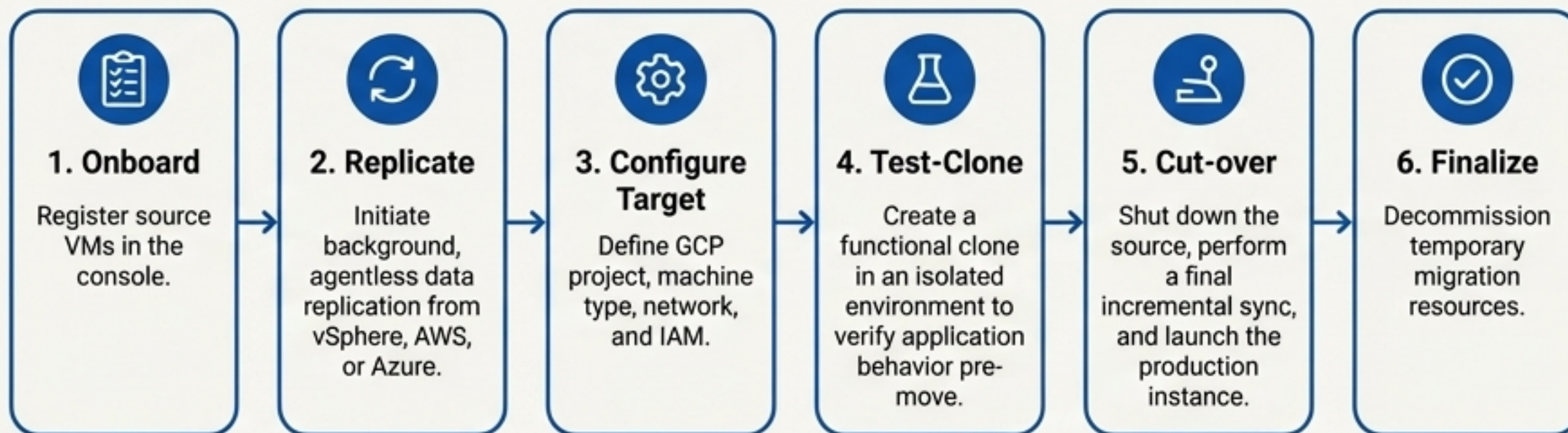
A Migration Factory for Large-Scale, Low-Downtime Transitions

A successful migration begins with a comprehensive assessment to map dependencies, followed by a phased execution that prioritizes validation and risk mitigation.

Assess: Migration Center

- Automated discovery of servers, databases, and network configurations.
- Advanced application dependency mapping to plan migration waves and prevent functional breakage.
- Integration with Looker Studio and CAST for deep-dive reporting on cloud readiness.

Execute: Migrate to Virtual Machines



Key Feature: Continuous block-level replication keeps the final cut-over downtime to a minimum, often measured in minutes.



Pillar 1: The Modernized Core

Achieving Business Continuity with Ransomware-Resistant Data Protection

Effective disaster recovery (DR) is governed by two metrics: Recovery Point Objective (RPO), the maximum acceptable data loss, and Recovery Time Objective (RTO), the time required to restore service. Google’s Backup and DR Service is a centralized, managed solution to meet stringent RPO/RTO targets.



“Incremental-Forever” Backups

After an initial full backup, only changed data blocks are captured, enabling frequent backups (e.g., database logs every 15 minutes) with minimal production impact.



Immutable Backup Vaults

Backups are committed to a Google-managed, isolated vault. They cannot be modified or deleted—even by admin accounts—until the retention period expires, providing a vital defense against ransomware.



“Mount and Migrate” Recovery

Near-instant recovery for databases (Oracle, SAP HANA) by mounting backup images directly as volumes, bypassing time-consuming restores.

Recovery Strategy Comparison

Recovery Strategy	Typical RTO	Typical RPO	Relative Cost
Cold (Backup & Restore)	24+ Hours	12-24 Hours	Lowest
Warm Standby	1-4 Hours	15-60 Minutes	Moderate
Hot (Active-Active)	< 5 Minutes	Near-zero	Highest

Automating Infrastructure Operations with Managed, Serverless Scheduling

The Challenge: Managing decentralized, system-level cron jobs creates operational overhead, lacks centralized visibility, and is prone to silent failures.

The Google Solution: Cloud Scheduler

-  A fully managed cron job service for automating batch jobs, big data operations, and infrastructure tasks.
- **Familiar & Flexible Syntax:** Supports standard unix-cron (e.g., 0 1 * * 0 for 1:00 AM every Sunday) and App Engine syntax.
-  **Built-in Resiliency:** Automatically retries failed jobs with configurable backoff durations and retry attempts. Integrates with Cloud Logging for a full audit trail.
- **Granular Control:** Pause, resume, and edit jobs without deleting the configuration, essential during maintenance windows.

Cloud Scheduler: Versatile Target Support

Target Type	Ingress Limitation	Typical Application
HTTP/S Endpoints	Configurable (Internal/External)	Triggering RESTful APIs or Webhooks
Cloud Pub/Sub	Internal	Orchestrating asynchronous microservices
App Engine HTTP	Project-Restricted	Managing legacy web application tasks
Cloud Run Functions	Internal/Restricted	Executing serverless administrative logic



Pillar 2: The Automated Engine

Connecting Disparate Applications to Eliminate Data Silos

The Challenge: Data fragmented across dozens of SaaS platforms (Salesforce, SAP) and internal databases creates **inefficiency** and prevents a unified view of the business.

The Google Solution: Application Integration (iPaaS)

- A serverless, fully managed platform to design, create, and manage integrations.
- **Low-Code Editor:** Intuitive, drag-and-drop interface allows business analysts and developers to build complex workflows.
- **Pre-built Connectivity:** Over 90 pre-built connectors for Google Cloud services and third-party SaaS applications like Salesforce, SAP, and ServiceNow.

Example Workflow: Salesforce to BigQuery Synchronization



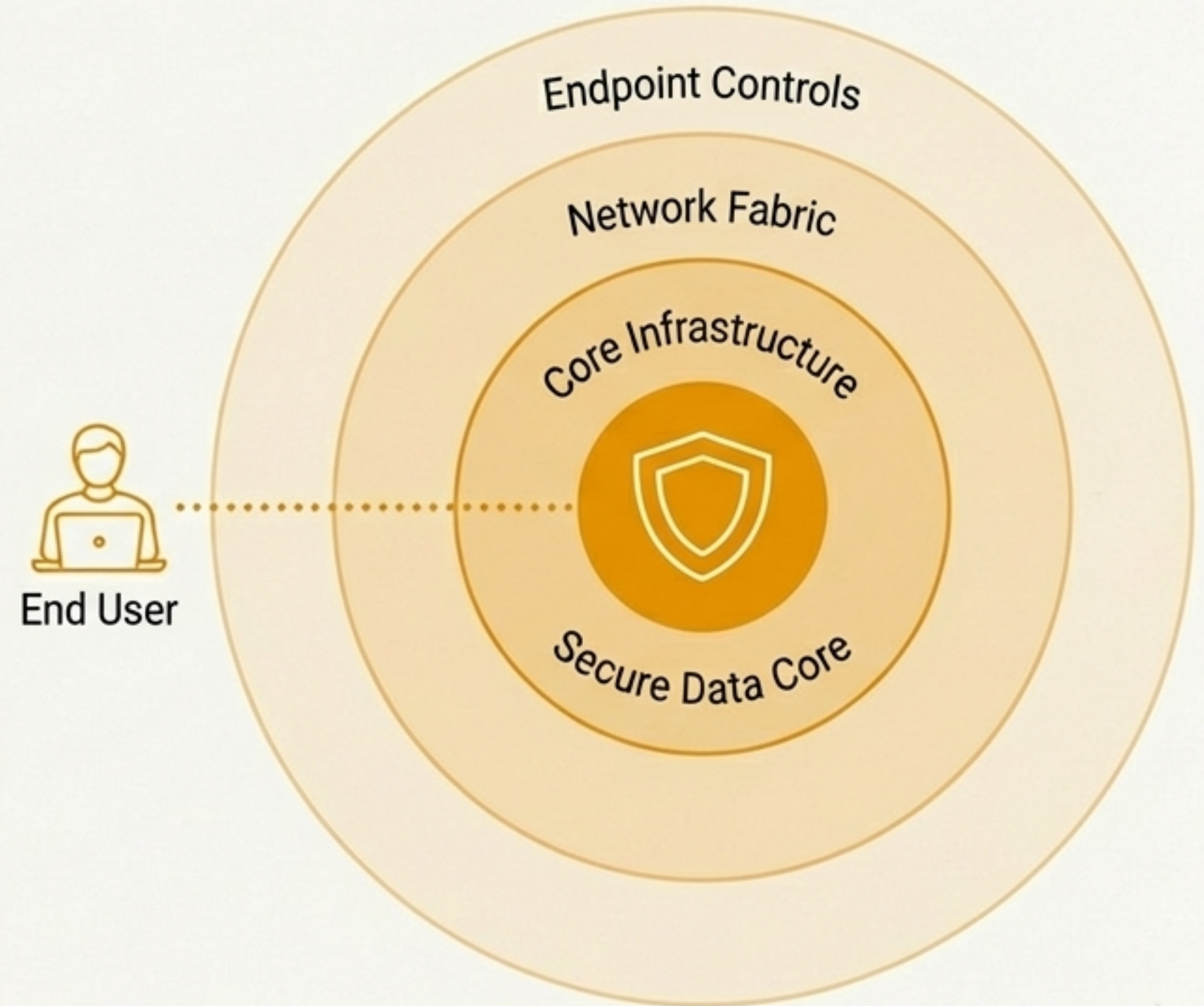


Pillar 3: The Secure Perimeter

Securing the Enterprise from the Browser to the Core Network

The Challenge: In a world of hybrid work, BYOD, and cloud-native applications, the traditional network perimeter has vanished. Security must be enforced at every layer, starting with the primary work interface—the browser—and extending through the network fabric.

The Google Solution: An integrated security model that provides centralized management and Zero Trust controls for endpoints, enforces granular policy on the network, and scales connectivity without compromising security.





Pillar 3: The Secure Perimeter

The Browser as the Enterprise Security Control Point

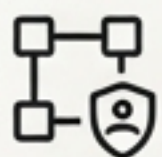
Chrome Enterprise extends the browser your teams already use into a manageable, secure endpoint for all web-based work, on any device.

Chrome Enterprise Core (\$0)	Chrome Enterprise Premium (\$6/user/month)
Centralized Management	All Core Features, Plus:
✓ Cloud-based policy management (>500 policies) ✓ Browser & extension version reporting ✓ Extension request & approval workflows ✓ Integration with UEM (Intune, VMware, JAMF)	Advanced Threat Protection ✓ Real-time Safe Browsing & deep malware scanning
	Data Loss Prevention (DLP) ✓ Prevent sensitive data uploads & password reuse
	Zero Trust Access Control ✓ Context-aware access for SaaS & private web apps (based on user, location, device)
	URL Filtering & Reporting ✓ Restrict URLs by category ✓ Security insights & evidence locker for investigation



Pillar 3: The Secure Perimeter

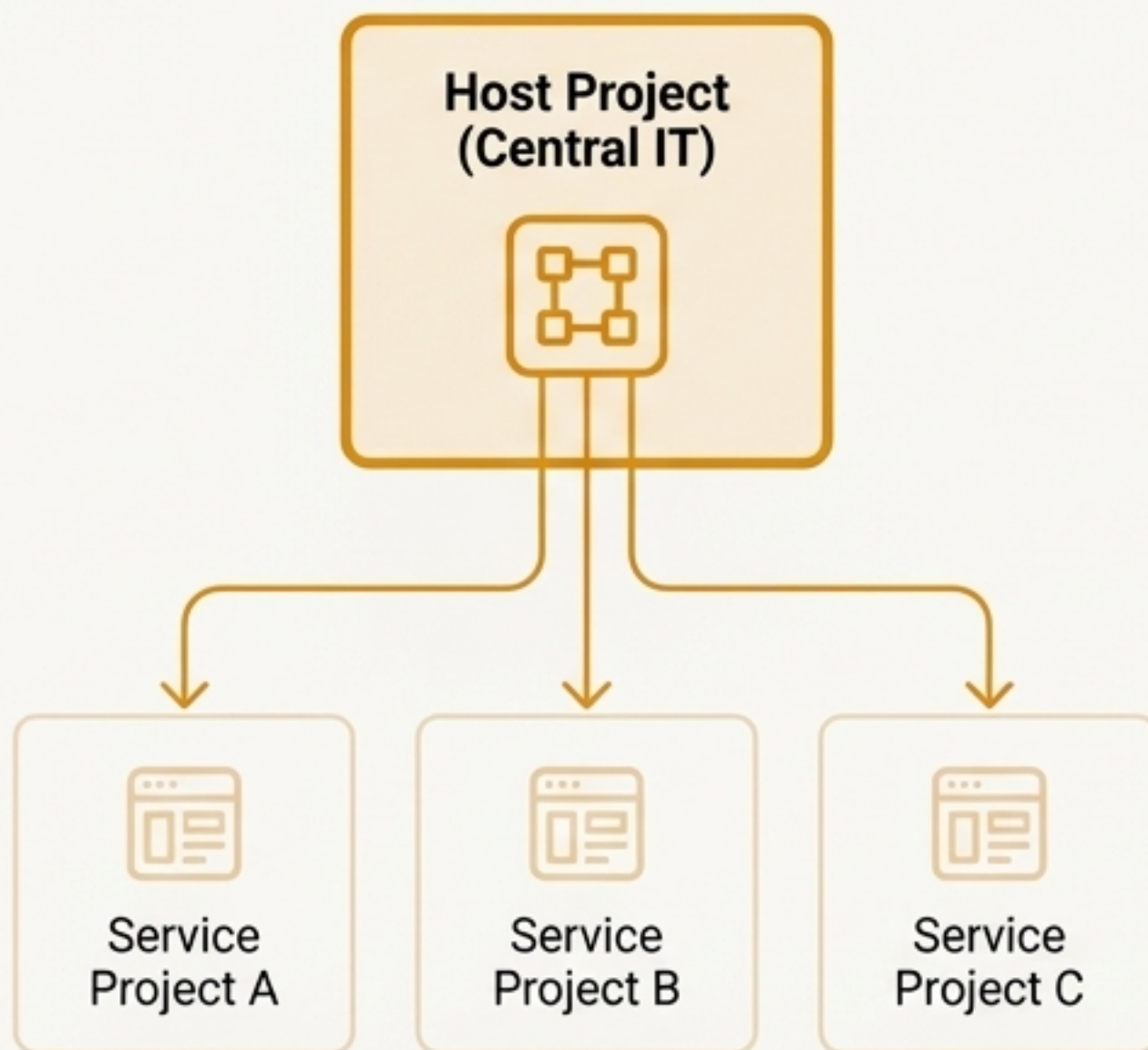
Centralized Governance with a Shared Network Foundation



Shared VPC: Principle of Least Privilege

A centralized networking team manages a "host project" containing the VPC, subnets, and firewalls. Application teams are granted "Network User" roles for specific subnets in their "service projects."

Benefit: Enforces consistent security posture at the organization level while giving developers autonomy to deploy workloads into pre-approved, secure network segments.



Hybrid Subnets: Seamless IP Preservation

Create a single logical subnet spanning both on-premises and Google Cloud VPC. Allows workloads to be migrated without changing their IP addresses, avoiding the most common cause of application downtime during a move.

Mechanism: Uses Cloud Router and longest-prefix matching to transparently direct traffic to the VM's current location.



Pillar 3: The Secure Perimeter

Scaling Global Connectivity with a Resilient Hub-and-Spoke Model

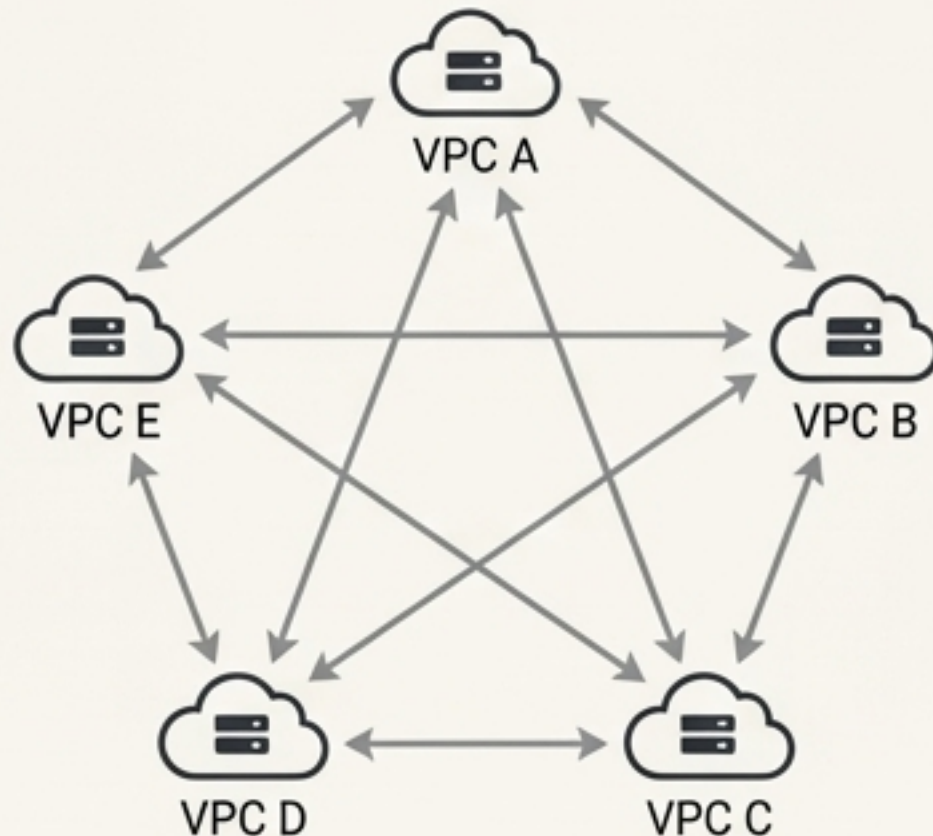
The Challenge

Traditional VPC peering is non-transitive, operationally complex to manage at scale, and subject to quotas, creating a fragile mesh.

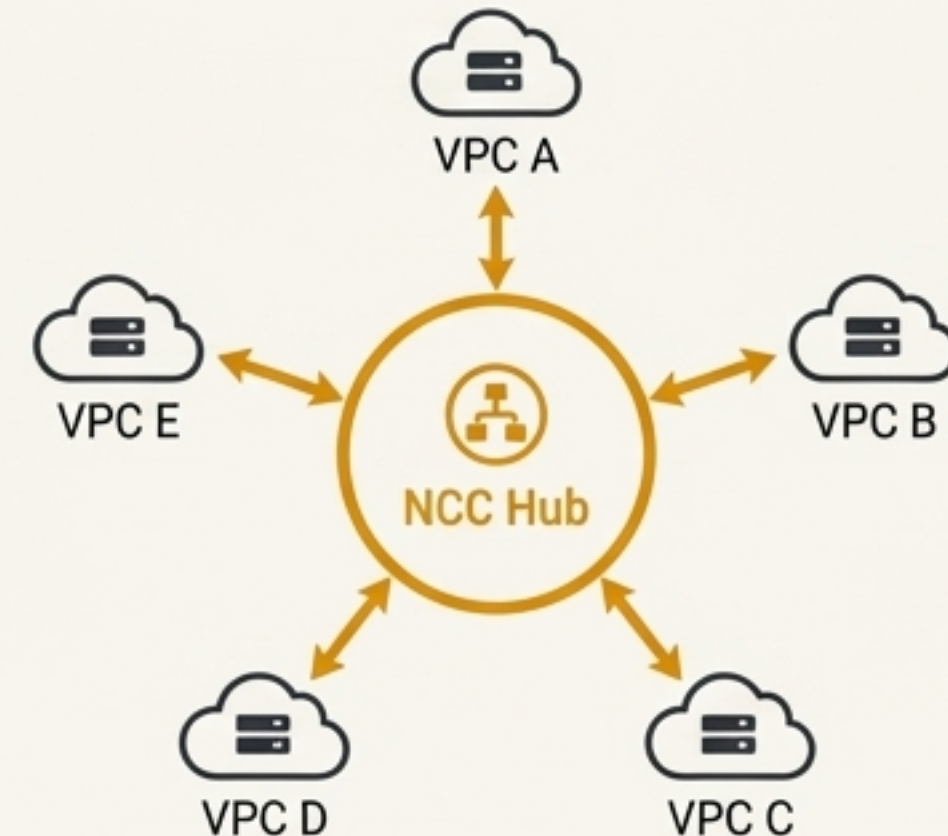
The Google Solution: Network Connectivity Center (NCC)

- A centralized hub for managing connectivity across VPCs and on-premises environments.
- **Massive Scale:** Supports up to 250 VPC spokes per hub and millions of VM instances.
- **Transitive Routing:** Spokes can communicate with each other through the hub, eliminating the need for a full mesh of peering links.
- **Architected for Resilience:** Built on three decoupled planes (Management, Control, Data). Its fail-static behavior ensures that if the control plane has an issue, the data plane continues forwarding traffic based on the last known good configuration.

Before: VPC Peering



After: Network Connectivity Center





Pillar 4: The Foundation of Trust

Enabling Developer Velocity with High-Fidelity Local Emulation

The Challenge

Developers need to build and test applications that interact with cloud services without incurring cloud costs, requiring constant connectivity, or dealing with inconsistent local environments.

The Google Solution: Local Service Emulators

High-fidelity local representations of GCP services that run on a developer's machine.

Benefits:

- Accelerates build-test-debug cycles.
- Enables offline development and integration testing.
- Eliminates cloud costs during development.
- Ensures a consistent, "run-ready" environment for every team member via setup scripts (e.g., ``setup-emulators.sh``).

Key Available Emulators

Emulator Service	gcloud Component ID	Primary Use Case
Cloud Pub/Sub	pubsub-emulator	Testing event-driven messaging flows
Cloud Bigtable	bigtable	Validating high-throughput NoSQL schemas
Cloud Spanner	cloud-spanner-emulator	Testing relational transactions at scale
Cloud Firestore	cloud-firestore-emulator	Developing mobile & web app backends



Pillar 4: The Foundation of Trust

Verifying Security and Operations Through Independent Audits

Trust must be earned and verified. Google Cloud's infrastructure and operations are regularly audited by independent third parties against the industry's most rigorous standards.

Focus on: SOC 2 Compliance

What it is

An audit based on the AICPA's five Trust Services Criteria: security, availability, processing integrity, confidentiality, and privacy.

Why it matters

A **SOC 2 Type II** report evaluates the *operational effectiveness* of controls over a period of time, providing deep assurance that security measures are consistently applied.

Google's Commitment

- Regular audits performed by objective firms like Ernst & Young LLP.
- Quarterly reports are available via the Compliance Reports Manager.
- "Bridge Letters" are issued between audits to summarize any material changes, ensuring continuous transparency.



The Cohesive Platform for Enterprise Transformation

Building a modern enterprise requires more than a collection of services; it requires an integrated ecosystem. Google provides a single platform where the journey to a **Modernized Core** is de-risked, the **Automated Engine** reduces toil, the **Secure Perimeter** extends from every user to the cloud, and the entire system rests on a verifiable **Foundation of Trust**.



This is the definitive platform for enterprise transformation.